

4. Developing Single Page application using Ojet navigation (using layout Components)

4.1. Oracle JET Single Page Application – Task-Manager

Introduction

This activity demonstrates how to create a fully functional **Task Manager** application using **Oracle JET** with the **MVVM architecture**, **Knockout.js observables**, and a rich set of Oracle JET UI components.

Learners will build a responsive single-page task management interface that supports:

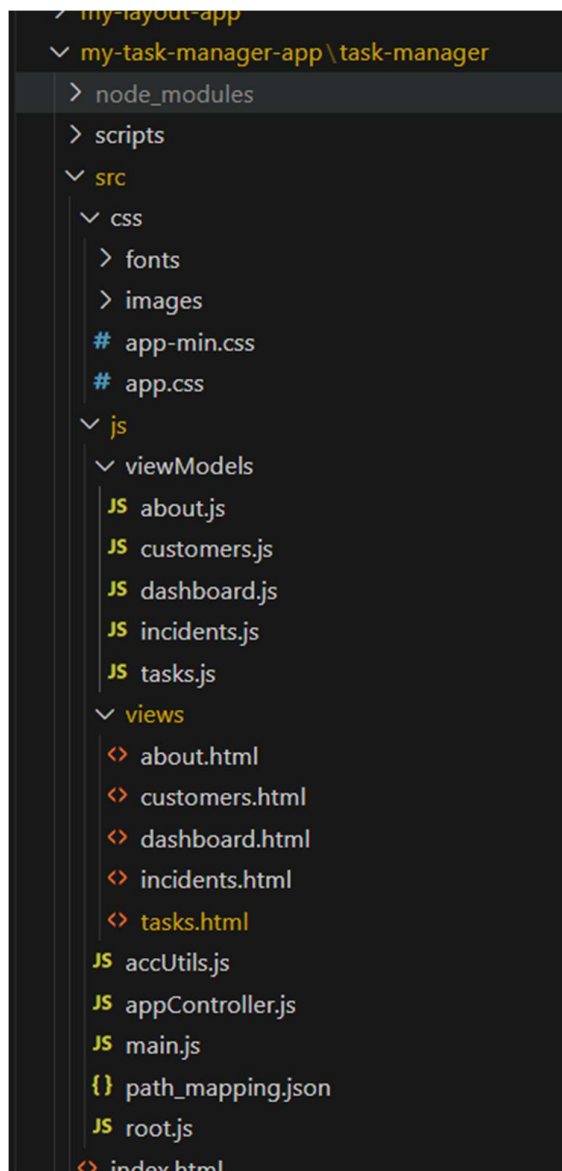
- Adding new tasks with optional due dates
- Filtering tasks by status (All, Pending, Completed)
- Marking tasks as completed
- Deleting tasks
- Dynamically rendering tasks using <oj-list-view> and ArrayDataProvider

This example follows Oracle JET best practices and showcases seamless data binding, event handling, and UI responsiveness, all within the context of a real-world use case.

Pre-requisite:

- Basic understanding of **Oracle JET architecture** (especially MVVM)
- Familiarity with **Knockout.js** bindings (observable, computed)
- Knowledge of how to use **JET components** like oj-input-text, oj-button, oj-list-view, oj-checkbox, etc.

Project structure:



1. Create a new folder named "my-task-manager-app" and move to that directory
2. In the command line run,
Ojet create task-manager - -tempalate=navdrawer
3. Then run ojet build web and ojet serve respectively (like we did before)
4. Create folders /js/viewModels/tasks.js and /js/views/task.html
5. Add the following code to tasks.js

my-task-manager-app > task-manager > src > js > viewModels > JS tasks.js > define() callback > TaskViewModel

```
1  define([
2    'knockout',
3    'ojs/ojarraydataprovider',
4    'ojs/ojinputtext',
5
6    'ojs/ojdatetimepicker',
7    'ojs/ojbutton',
8    'ojs/ojlistview',
9    'ojs/ojlistitemlayout',
10   'ojs/ojcheckboxset',
11   'ojs/ojradio',
12   'ojs/ojoption',
13   'ojs/ojlabel'
14 ], function (ko, ArrayDataProvider) {
15
16   function TaskViewModel() {
17     const self = this;
18
19     self.newTask = ko.observable('');
20     self.newTaskDue = ko.observable('');
21     self.filterType = ko.observable('all');
22
23     self.tasks = ko.observableArray([]);
24
25     self.filteredTasks = ko.computed(() => {
26       switch (self.filterType()) {
27         case 'pending':
28           return self.tasks().filter(task => !task.done);
29         case 'completed':
30           return self.tasks().filter(task => task.done);
```

```
31       default:
32         return self.tasks();
33       }
34     });
35
36     self.filteredTasksDP = ko.computed(() => {
37       return new ArrayDataProvider(self.filteredTasks(), { keyAttributes: 'id' });
38     });
39
40     self.addTask = () => {
41       const title = self.newTask().trim();
42       const due = self.newTaskDue();
43       if (title) {
44         self.tasks.push({
45           id: Date.now().toString(),
46           title: title,
47           due: due || 'No date',
48           done: false
49         });
50         self.newTask('');
51         self.newTaskDue('');
52       }
53     };
54
```

6. Add following code to task-manager.html file

```

tasks.html 6 X
my-task-manager-app > task-manager > src > js > views > tasks.html > oj-panel.oj-sm-padding-4x.oj-bg-neutral-10.oj-rounded > div.oj-flex.oj-sm-paddin
1 <oj-panel class="oj-sm-padding-4x oj-bg-neutral-10 oj-rounded">
2   <div class="oj-flex oj-sm-padding-4x-around">
3     <div class="oj-flex-item oj-sm-12 oj-md-8 oj-lg-6">
4
5       <h2>Task Manager (<span data-bind="text: filteredTasks().length"></span>)</h2>
6
7       <!-- Filters -->
8       <oj-radioset value="{{filterType}}" class="oj-sm-margin-y-2">
9         <oj-option value="all">All</oj-option>
10        <oj-option value="pending">Pending</oj-option>
11        <oj-option value="completed">Completed</oj-option>
12      </oj-radioset>
13
14      <!-- New Task Input -->
15      <oj-input-text label-hint="New Task" value="{{newTask}}" class="oj-form-control-max-width-md"></oj-input-text>
16
17      <!-- Due Date -->
18      <oj-input-date label-hint="Due Date" value="{{newTaskDue}}" class="oj-sm-margin-top-2"></oj-input-date>
19
20      <!-- Add Button -->
21      <oj-button on-oj-action="[[addTask]]" class="oj-sm-margin-top-2">Add Task</oj-button>
22
23      <!-- Task List -->
24      <oj-list-view data="[[filteredTasksDP]]" class="oj-sm-margin-top-4" selection-mode="none">
25        <template slot="itemTemplate">
26          <oj-list-item-layout>
27            <div class="oj-flex oj-align-items-center">
28
29              <!-- Checkbox for Done -->
30              <oj-checkbox
31                value="{{current.data.done}}"
32                on-value-changed="[[({}) => toggleDone($current.data)]]">
33            </oj-checkbox>
34
35            <!-- Title & Due Date -->
36            <div class="oj-sm-margin-start-2">
37              <span data-bind="text: $current.data.title, css: { 'oj-text-line-through': $current.data.done }"></span>
38              <div data-bind="text: 'Due: ' + $current.data.due"></div>
39            </div>
40
41            <!-- Delete -->
42            <oj-button class="oj-sm-margin-start-auto" on-oj-action="[[({}) => removeTask($current.data)]]" chroming="borderless" display="icons">
43              <span slot="startIcon" class="oj-ux-ico-trash"></span>
44            </oj-button>
45          </div>
46        </oj-list-item-layout>
47      </template>
48    </oj-list-view>
49  </div>
50</div>
51</div>
52</div>
53</oj-panel>

```

7. add the route to the app in “appcontroller.js” file (js/appcontroller.js)

```

let navData = [
  { path: '', redirect: 'dashboard' },
  { path: 'dashboard', detail: { label: 'Dashboard', iconClass: 'oj-ux-ico-bar-chart' } },
  { path: 'tasks', detail: { label: 'Tasks', iconClass: 'oj-ux-ico-list' } },
  { path: 'incidents', detail: { label: 'Incidents', iconClass: 'oj-ux-ico-fire' } },
  { path: 'customers', detail: { label: 'Customers', iconClass: 'oj-ux-ico-contact-group' } },
  { path: 'about', detail: { label: 'About', iconClass: 'oj-ux-ico-information-s' } }
];

```

8. Run ojet serve on command terminal

9. Navigate to the tasks page in the browser.

ORACLE[®] App Name

john.hancock@oracle.com

Dashboard

Tasks

Incidents

Customers

About

Task Manager (0)

All

Pending

Completed

New Task

Due Date

Add Task

No items to display.

11. Add task and give the due date for its completion

12. You will see it gets added in the list

Task Manager (1)

All

Pending

Completed

New Task

Due Date

Add Task

demo task

Due: 2025-06-20

You can then see the task in all and pending list too and (2) along with heading task manager means there are two tasks

Task Manager (2)

☐ All

☒ Pending

☐ Completed

New Task

Due Date



Add Task

demo task

Due: 2025-06-20



demo task 2

Due: 2025-06-21



Congratulations !! you have created a small task manager application!!